

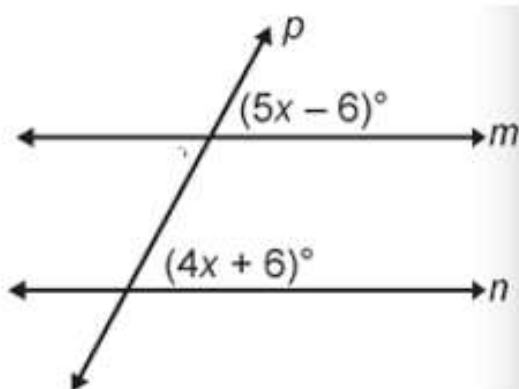
Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Mathematics

1. A shopkeeper marks the prices of items to the nearest ten rupees. If a shirt is priced at ₹475, how much will a customer likely pay if the shopkeeper follows this practice?
(A) ₹475 (B) ₹480
(C) ₹500 (D) ₹400
2. In the given figure, **lines m and n are parallel**, and **p is a transversal**. The angles marked are $(5x-6)^\circ$ on line **m** and $(4x+6)^\circ$ on line **n**.

What is the value of **x**?



- (A) 8 (B) 10
(C) 12 (D) 14

3. **Statement I:** A regular hexagon has 6 lines of symmetry.
Statement II: A rectangle has only 2 lines of symmetry.
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true and Statement II is false
(D) Statement I is false and Statement II is true
4. The frequency table shows the number of hours students' study daily:

Hours	Number of Students
1	5
2	8
3	12
4	10

How many students study for more than 3 hours daily?

- (A) 10 (B) 15
(C) 20 (D) 25

5. The pictograph below shows the number of ice creams sold in a week at a shop.

Each  represents 10 ice creams.

Days & Ice Creams Sold

Monday:    

Tuesday:   

Wednesday:     

Thursday:  

Friday:      

Saturday:       

Sunday:     

How many more ice creams were sold on Saturday than on Wednesday?

- (A) 10 (B) 20
(C) 30 (D) 40
6. A librarian arranged five books with page counts: 245, 1024, 3567, 89, and 12458. **Which book has the second-highest number of pages?**
- (A) 245 (B) 1024
(C) 3567 (D) 12458
7. A playground is in the shape of a rectangle with length 25 m and breadth 20 m. In the centre, there is a circular fountain of radius 3 m. **What is the area of the playground excluding the fountain?**
- (A) 471.74 m^2 (B) 500 m^2
(C) 471.86 m^2 (D) 528.14 m^2

8. A designer creates a logo that looks exactly the same when rotated 90° , 180° , and 270° . **How many lines of symmetry does it have?**

(A) 2 (B) 4
(C) 6 (D) 8

9. **Arrange in ascending order:** $-12, 5, -8, 0, -3, 7$

(A) $-12, -8, -3, 0, 5, 7$
(B) $7, 5, 0, -3, -8, -12$
(C) $-12, -8, 0, -3, 5, 7$
(D) $-3, -8, -12, 0, 5, 7$

10. In a reflection activity, Raj places a mirror along the side of the letter "M". The reflection shows the same letter.

Which of these lines is NOT a line of symmetry for "M"?

(A) Vertical (B) Horizontal
(C) Diagonal left (D) Diagonal right

11. A bus starts with 25 passengers. At the first stop, 8 get down and 12 get in. At the next stop, 5 get down and 7 get in.

How many are now in the bus?

(A) 31 (B) 32
(C) 33 (D) 34

12. The perimeter of a square is equal to that of a rectangle with length 18 cm and breadth 10 cm. **What is the area of the square?**

(A) 144 cm^2 (B) 169 cm^2
(C) 196 cm^2 (D) 225 cm^2

13. The pictograph shows the number of pencils sold in a shop over 5 days. Each

 represents 5 pencils.

Mon: 

Tue: 

Wed: 

Thu: 

Fri: 

How many pencils were sold from Tuesday to Thursday?

- (A) 60 (B) 70
(C) 75 (D) 80

14. In a city's voter list, numbers were written without commas. A person reads the number 50341678. How would it be expressed in the Indian numeration system with commas?

- (A) 5,03,41,678 (B) 50,341,678
(C) 5,034,167,8 (D) 50,34,16,78

15. Riya wants to fence her square-shaped plot. If the cost of fencing is ₹25 per meter and she pays ₹800, what is the side length of her plot?

- (A) 6 m (B) 8 m
(C) 10 m (D) 12 m

16. In a game, scoring a goal gives +7 points, a foul give -4 points, and a save gives +3. If a player has 2 goals, 3 fouls, and 5 saves, what is the total?

- (A) 12 (B) 17
(C) 20 (D) 23

17. A rectangular garden is 12 m long and 8 m wide. If a 1 m wide path is built inside along its boundary, what is the area of the path?

- (A) 36 m² (B) 40 m²
(C) 44 m² (D) 48 m²

18. Which property is illustrated by the statement: $15 \times (10 + 3) = (15 \times 10) + (15 \times 3)$?

- (A) Commutative (B) Associative
(C) Distributive (D) Identity

19. Match the type of angle with its correct measure range from the options given:

Column I (Angle Type)	Column II (Measure Range)
(a) Acute angle	(P) Exactly 90°
(b) Obtuse angle	(Q) More than 180° but less than 360°
(c) Right angle	(R) More than 0° but less than 90°
(d) Reflex angle	(S) More than 90° but less than 180°

- (A) (a)–(s), (b)–(p), (c)–(q), (d)–(r)
(B) (a)–(r), (b)–(s), (c)–(p), (d)–(q)
(C) (a)–(p), (b)–(r), (c)–(s), (d)–(q)
(D) (a)–(q), (b)–(r), (c)–(s), (d)–(p)

20. A teacher asks students to form the smallest 4-digit number using digits 7, 0, 4, 9 without repetition. What will it be?

- (A) 0479 (B) 4079
(C) 7049 (D) 9047

- 21.** In a survey of favourite fruits, 30% chose apples, $\frac{1}{4}$ chose bananas, 0.2 chose grapes, and the rest chose oranges. If 90 students chose oranges, **how many students were surveyed in total?**
- (A) 400 (B) 360
(C) 300 (D) 450
- 22.** Two traffic lights change after every 45 seconds and 60 seconds respectively. They turned green together at 8:00 AM. **When will they next turn green together?**
- (A) 8:03 AM (B) 8:04 AM
(C) 8:05 AM (D) 8:06 AM
- 23.** Aisha is designing a logo for her school competition. She draws a shape and folds it along a line. Both halves match perfectly. **Which of the following figures could she have folded if the line was diagonal?**
- (A) A scalene triangle
(B) An isosceles right triangle
(C) An irregular pentagon
(D) A letter 'S'
- 24.** A school parade has students in rows of 12, 15, or 18. What is the smallest number of students needed so that they can be arranged in any of these row sizes without leaving anyone out?
- (A) 90 (B) 120
(C) 180 (D) 360
- 25.** Which number is exactly midway between 3.18 and 3.24?
- (A) 3.20 (B) 3.21
(C) 3.22 (D) 3.23
- 26.** The following pictograph shows the number of trees planted by four classes in a week. If each  represents 8 trees and Class 4 planted 6 symbols plus half a symbol, how many more trees did Class 4 plant than Class 1, which has 4 symbols?
- (A) 12 (B) 16
(C) 20 (D) 24
- 27.** A farmer fences a triangular plot. Two angles measure 65° and 45° . **What is the measure of the third angle?**
- (A) 60° (B) 70°
(C) 80° (D) 90°
- 28.** A number is divisible by both 9 and 11. **Which of the following could be the number?**
- (A) 1089 (B) 1001
(C) 1221 (D) 1331
- 29.** Rohan cuts out a paper snowflake by folding a circular piece 4 times and cutting shapes along the edges. When he unfolds it completely, how many lines of symmetry will the snowflake most likely have?
- (A) 2 (B) 4
(C) 8 (D) Infinite

30. Two straight roads intersect at a point, forming four angles. If one angle is **75°**, **what is the measure of its vertically opposite angle?**

- (A) 75° (B) 105°
(C) 15° (D) 285°

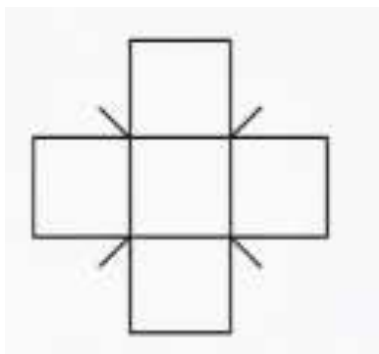
31. A vertical cut through the middle of a cube will produce which cross-section?

- (A) Square (B) Rectangle
(C) Triangle (D) Circle

32. The HCF of two numbers is 14 and their LCM is 280. If one number is 56, **what is the other?**

- (A) 70 (B) 84
(C) 98 (D) 112

33. Look at the 2D net shown below:



Which 3D shape will be formed by folding this net?

- (A) Triangular prism (B) Cuboid
(C) Square pyramid (D) Cylinder

34. How many small cubes of side 2 cm can be packed into a box of dimensions 10 cm × 8 cm × 6 cm?

- (A) 60 (B) 120
(C) 240 (D) 480

35. In a quadrilateral, three angles are 85°, 95°, and 110°.

What is the fourth angle?

- (A) 60°
(B) 70°
(C) 80°
(D) 90°

36. In a kaleidoscope, two mirrors are joined at 60°.

How many lines of symmetry does the pattern seen through it have?

- (A) 2 (B) 3
(C) 6 (D) 12

37. Which of the following nets can form a closed cube?

- (A) Four squares in a row
(B) A cross of six squares
(C) Three squares in an L-shape
(D) Two squares stacked

38. Reema used 0.6 of her money for a book and $\frac{3}{10}$ for a pen.

What fraction of her money is left?

- (A) $\frac{1}{5}$ (B) $\frac{1}{10}$
(C) $\frac{2}{5}$ (D) $\frac{3}{10}$

39. If a line segment AB is 8 cm long and C is a point between A and B such that AC = 3 cm, what is CB?

- (A) 3 cm (B) 5 cm
(C) 8 cm (D) 11 cm

40. A submarine is 350 m below sea level. It rises 120 m, then descends 85 m, and finally rises 60 m. **What is its final position relative to sea level?**
- (A) 165 m below (B) 255 m below
(C) 255 m above (D) 165 m above
41. Which is the **smallest**?
- (A) $\frac{5}{6}$ (B) 0.83
(C) $\frac{7}{8}$ (D) 0.92
42. **The range of data:** 14, 22, 9, 25, 31, 17 is:
- (A) 22 (B) 23
(C) 24 (D) 25
43. If a solid has 6 faces, 12 edges, and 8 vertices, **which shape is it?**
- (A) Square pyramid (B) Cylinder
(C) Triangular prism (D) Cuboid
44. A wire bent into a circle of radius 7 cm is reshaped into a square. **What is the side of the square?**
- (A) 10 cm (B) 11 cm
(C) 12 cm (D) 13 cm
45. The temperature at 6 a.m. was -4°C . It increased by 7°C by noon, dropped by 9°C in the evening, and rose by 5°C at night. **What was the final temperature?**
- (A) -1°C (B) -2°C
(C) 1°C (D) 0°C
46. A rope 12.75 m long is cut into 5 equal pieces. One piece is then cut into 3 equal parts. **What is the length of each smallest part?**
- (A) 0.85 m (B) 0.75 m
(C) 0.95 m (D) 0.65 m
47. In a quiz, each correct answer scores +5, each wrong answer scores -3 , and passing a question gives 0. If Ravi answered 12 correctly, 5 wrongly, and passed 3, **what is his total score?**
- (A) 45 (B) 39
(C) 51 (D) 33
48. The mean of 5 numbers is 18. If four numbers are 15, 20, 22, and 12, **what is the fifth?**
- (A) 21 (B) 22
(C) 23 (D) 24
49. During a sale, the price of a shirt was reduced by $\frac{2}{5}$ of its original price. If the sale price is ₹450, **what was the original price?**
- (A) ₹750 (B) ₹700
(C) ₹650 (D) ₹800
50. A room 5 m $\times$ 4 m is to be covered with tiles of 25 cm $\times$ 20 cm. **How many tiles are needed?**
- (A) 400 (B) 420
(C) 440 (D) 460

